10/21/2026

ACADEMIC REFLECTIVE JOURNAL

L07 MODEL EVALUATION

1. INTRODUCTION – (ZAYAAN)

In this lab, we assessed a machine learning model using the Titanic dataset to predict passenger survival. Rather than just looking at the accuracy of our model, we examined its performance in more detail by calculating the confusion matrix, precision, recall, and F1-score. In addition, we explored cross-validation to evaluate the model's consistency across different data splits. This task was completed, as demonstrated in the attached file.

2. DESCRIPTION OF EXPIRIENCE/TOPIC - (JENNIFER)

In this experience accuracy shows how to measure a model with a correct predictions but also showing the types of mistakes it can possibly if not looked at properly. This lab really shows the strategies and tools to carefully evaluate the methods such as the dataset being divided into the training examples and testing samples which is where we use the confusion matrix.

3. PERSONAL REFLECTION – (SAFA BASIL)

I learned that accuracy does not explain every kind of prediction error. Precision and recall help answer different questions about model performance, and the F1-score balances both. Cross-validation also helps show whether a model performs consistently across different data splits.

4. DISCUSSION OF IMPROVEMENT AND LEARNING - (JENNIFER)

This lab made me more cautious on how to interpret the data when receiving the results and to look more beyond the single type of accuracy and the type of mistakes is what matter most when a problem is being solve. By being able to train the logistics in regressing the models and using python tools. Learning how to create a confusion matrix and understanding how to read the chart and a better understanding of false positives and false negatives.

5. CONCLUSION - (JENNIFER)

Overall, lab 7 has shown me about the confusion matrices, F1-score, cross validation, and precision with the different model performances. While depending on the data that we have gotten it varies depending on the performances and data that was used in this evaluation. By evaluating these models, it requires a lot more of accuracy. This also gives us an understanding all the types of errors and by checking the performances between all the splits to help make a better judgement and validations.